

Here’s your complete **100-mark printable test paper** for Class XI Commerce Applied Mathematics, covering:

- **Chapter 1: Number Systems (19 marks)**
- **Chapter 2: Indices & Logarithms (81 marks)**
- **Total Time:** 60 minutes
- **Includes full solutions and justifications**

 Combined Chapter Test

Class: XI Commerce **Subject:** Applied Mathematics **Total Marks:** 100 **Time:** 60 minutes **Instructions:** Attempt all questions. Show full working for medium and hard questions.

 Section A: Number Systems (19 marks)

 Easy (1 mark each)

1. Convert decimal 25 to binary.
2. Convert binary 10101 to decimal.

 Medium (3 marks each)

3. Add: $1101 + 1011$
4. Subtract: $10110 - 1101$
5. Multiply: 101×11

 Hard (5 marks each)

6. Divide: $11001 \div 11$
7. Convert decimal 156 to binary and verify by converting back to decimal.

 Section B: Indices & Logarithms (81 marks)

 Topic: Indices (27 marks)

Easy (1 mark)

1. Simplify: $2^3 \cdot 2^4$

Medium (3 marks each)

2. Solve for x : $5^x = 125$
3. Simplify: $\frac{a^5 \cdot a^{-2}}{a^3}$

Hard (5 marks each)

4. If $2^x = 8$ and $3^y = 27$, find $x + y$
5. Prove: $\frac{(x^2)^3}{x^4} = x^2$

 Topic: Logarithms (27 marks)

Easy (1 mark)

1. Find: $\log_{10} 1000$

Medium (3 marks each)

2. Solve: $\log_2 x = 5$
3. Simplify: $\log_5 25 + \log_5 5$

Hard (5 marks each)

4. Prove: $\log_a b + \log_b a = 2$
5. If $\log_{10} x = 2$ and $\log_{10} y = 3$, find $\log_{10} (x^2 y^3)$

 Topic: Antilogarithms (27 marks)

Easy (1 mark)

1. Find the antilog of 2 (base 10)

Medium (3 marks each)

2. Find x if $\log_{10} x = 2.3010$
3. Evaluate: $\text{antilog}_{10} (0.4771)$

Hard (5 marks each)

4. If $\log_{10} x = 3.69897$, find x
5. Show that $\text{antilog}_{10} (\log_{10} a + \log_{10} b) = ab$